Lakshya JEE 2.0 2025

Maths

DPP: 7

Relations and Functions

Q1 If $f(x)=\frac{2x-3}{5-7x}$, then find domain and range of $f(x)$.

Q2 Find maximum and minimum value of $f(x)=12\cos x-5\sin x$

Q3 If x is real, then the expression $\frac{x+2}{(2x^2+3x+6)}$ takes all values in the interval

(A) $\left(\frac{1}{13},\frac{1}{3}\right)$
(B) $\left[-\frac{1}{13},\frac{1}{3}\right]$
(C) $\left(-\frac{1}{3},\frac{1}{13}\right)$
(D) $\left(-\infty,-\frac{1}{13}\right]\cup\left[\frac{1}{3},\infty\right)$

Q4 The range of $f(x)=2+2x-x^2$ is

(A) $(-\infty,\infty)$
(B) $(-\infty,1)$
(C) $(-\infty,3]$
(D) $(0,\infty)$

Q5 Range of the function $f\left(x\right)=\frac{x^2+x+2}{x^2+x+1};x\in R$ is

(A) $(1,\infty)$
(B) $(1,11/7)$
(C) $(1,7/3]$
(D) $(1,7/5)$

Q6 The range of the function $f(x)=\frac{1}{2-\cos 3x}$ is

(A) $\left[\frac{-1}{3},0\right]$
(B) R
(C) $\left[\frac{1}{3},1\right]$
(D) None of these

Q7 Range of function $f(x)=\frac{x^2-3x+2}{x^2+x-6}$ is $R-\left\{\frac{a}{b},c\right\}$ What is $a+b+2c$?

(A) 4 (B) 5
(C) 6 (D) 8

Q8 Range of function $f(x)=\sqrt{2-x}+\sqrt{1+x}$ is $[a,b]$. What is the value of $a^2+b^2=$?

(A) 5 (B) 9
(C) 10 (D) 25

Q9 If $x\in[3,5)$, then find range of $f\left(x\right)=4x^2-28x+51$ [**Hint:** $f(x)=(2x-7)^2+2$ and then use elementary method]

Q10 Find range of $f(x)=9+12\cos x-\left(4\sin^2 x\right)$ $\left[\text{Hint: } f(x)=(2\cos x+3)^2-4\right]$

Q11 Find sum of smallest positive integer and largest negative integer belonging to range of $f\left(x\right)=\frac{x^2+1}{2x}$

Q12 If $A\in(-3,2]$ then $\frac{1}{A^2}\in\cdots$

Q13 If $A\in(-\infty,-7]\cup[3,6)$ Then $\frac{1}{A}\in$

Q14 Find Range of $f(x)=\sqrt{25-x^2}$

Q15

	Function		Range
(1)	$f(x) = \dfrac{x^2}{1+x^2}$	(P)	$R - \left\{\dfrac{-3}{2}\right\}$
(2)	$f(x) = \dfrac{3x-1}{5-2x}$	(Q)	$(-\infty,-1]\cup\left[\dfrac{1}{3},\infty\right)$
(3)	$f(x) = \dfrac{1}{2\sin x+1}$	(R)	$[3, 7]$
(4)	$f(x) = x^2 - 4x + 7$, $x \in (1,5]$	(S)	$[3, 12]$
(5)	$f(x) = \sqrt{3}\sin x - \cos x + 5$	(T)	$[0, 1)$

Q16 Range of $f(x) = \frac{3}{2-x^2}$ is
(A) $\left(-\infty, \frac{3}{2}\right]$
(B) $(-\infty, 0) \cup \left[\frac{3}{2}, \infty\right)$
(C) $(-\infty, 0] \cup \left[\frac{3}{2}, \infty\right)$
(D) $\left(-\infty, \frac{2}{3}\right]$

Q17 The range of the function $f(x) = \frac{1+x^2}{x^2}$ is equal to:
(A) $(0, 1)$
(B) $[0, 1]$
(C) $(1, \infty)$
(D) $[1, \infty)$

Q18 The range of the function $f(x) = \cos^2 x - 5\cos x - 9$ is:
(A) $[-13, 3]$ (B) $[0, 3]$
(C) $[-13, -3]$ (D) $[-13, -9]$

Q19 The range of the function $f\left(x\right) = \frac{1}{2-\sin 3x}$,
(A) $(1/3, 1)$
(B) $[1/3, 1]$
(C) $[1/3, 1)$
(D) None of these

Q20 Let $f(x) = \left(1+b^2\right)x^2 + 2bx + 1$ and let $m(b)$ be the minimum value of $\mathrm{f(x)}$. As b varies, the range of $m(b)$ is
(A) $[0, 1]$
(B) $\left[0, \frac{1}{2}\right]$
(C) $\left[\frac{1}{2}, 1\right]$
(D) $(0, 1]$

Q21 Find the range of $f(x) = \ln(x-1)$
(A) $(0, \infty)$
(B) $[2, \infty)$
(C) $(-\infty, \infty)$
(D) $[0, \infty)$

Q22 Find the range of $f(x) = e^x - 2$
(A) $(2, \infty)$
(B) $(1, \infty) x \in [-4, -\pi] \cup [0, \pi]$
(C) $(-2, \infty)$
(D) $(-1, \infty)$

Q23 Find the range of $f(x) = \frac{4}{3^x} - 2$.

Q24 The range of the function $f(x) = \log_5\left(25 - x^2\right)$, is
(A) $[0, 5]$ (B) $[0, 2)$
(C) $(0, 2)$ (D) none of these

Q25 The range of function $y = f(x)$ if $(34)^x + (34)^y = 34$ equals
(A) $(-\infty, -1)$
(B) $(-\infty, 1)$
(C) $(-\infty, 1]$
(D) $(1, \infty)$

Answer Key

Q1 Domain = R – $\left\{\frac{5}{7}\right\}$
Range = R – $\left\{-\frac{2}{7}\right\}$

Q2 maximum = 13, minimum = - 13

Q3 (B)

Q4 (C)

Q5 (C)

Q6 (C)

Q7 (D)

Q8 (B)

Q9 $f(x) \in [2, 11)$

Q10 $R_f \in [-3, 21]$

Q11 0

Q12 $\frac{1}{A^2} \in \left(\frac{1}{9}, \infty\right)$

Q13 $\frac{1}{A} \in \left[-\frac{1}{7}, 0\right) \cup \left(\frac{1}{6}, \frac{1}{3}\right]$

Q14 $R_f \in [0, 5]$

Q15 1-T, 2-P, 3-Q, 4-S, 5-R

Q16 (B)

Q17 (C)

Q18 (C)

Q19 (B)

Q20 (D)

Q21 (C)

Q22 (C)

Q23 range $\in (-2, \infty)$

Q24 (D)

Q25 (B)

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